

Deep Learning

CSE641/ECE555

Lecture 7 | Take your own notes during lectures

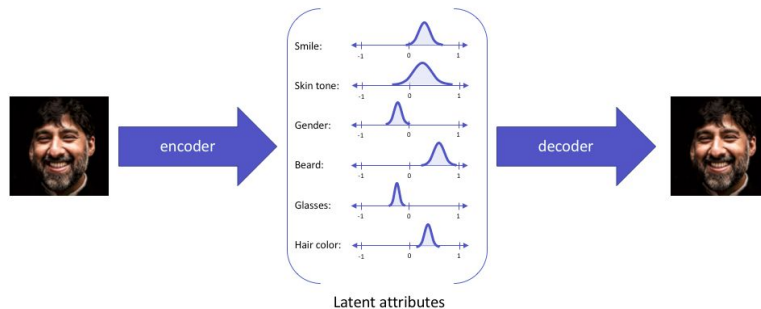
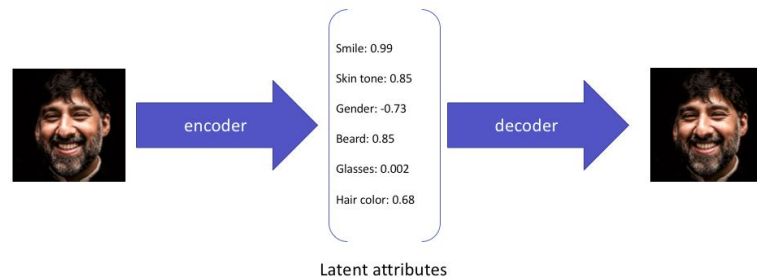
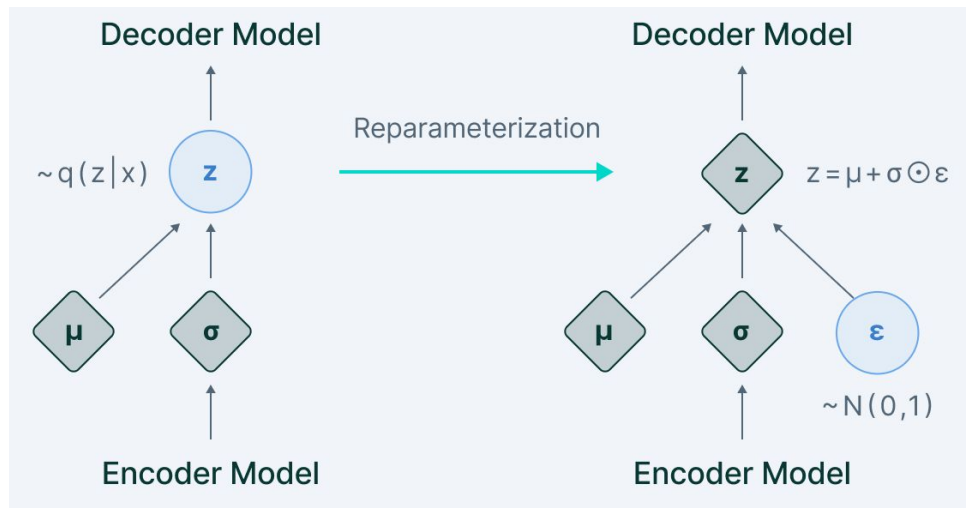
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INFORMATION TECHNOLOGY **DELHI**

Variational AutoEncoders (VAEs)

VAEs extend the deterministic latent space to probabilistic space by express their latent attributes as a probability distribution, leading to the formation of a continuous latent space that can be easily sampled and interpolated.



Variational AutoEncoders (VAEs)

Impose a distribution for Z and then train a decoder g so that $g(Z)$ matches the training data.

(Kingma and Welling, 2013)

Recall the maximum likelihood training objective

$$\max_{\theta} \mathbb{E}_{p_D(x)} [\log p_{\theta}(x)]$$

But in AE we generate X via $X \rightarrow Z \rightarrow X$
`.decode(Z) = decode(encode(X))`

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$$\max_{\theta} \mathbb{E}_{p_D(x)} [\log p_{\theta}(x)] = \mathbb{E}_{p_D(x)} [\log \int_z p_{\theta}(x|z) p(z) dz]$$

We can rewrite the objective using marginalization over 'Z'

Observe the quantity inside integral looks like our decoder
But this equation is intractable Why?

Variational AutoEncoders (VAEs)

Overcome intractability?

Assume a network (θ) that maps from Z to X (Decoder)

Assume a network (ϕ) to map from X to Z (Encoder)

$$p_{\theta}(x, z) = p_{\theta}(x|z)p(z)$$

$$q_{\phi}(x, z) = q_{\phi}(z|x)p_D(x)$$

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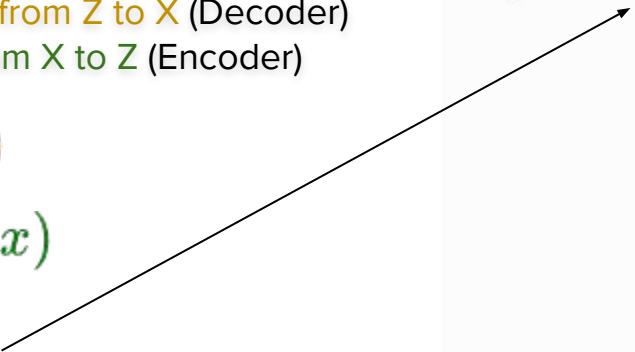
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$$\begin{aligned}\log \int_z p_{\theta}(x|z)p(z)dz &= \log \int_z \frac{q_{\phi}(z|x)}{q_{\phi}(z|x)} p_{\theta}(x|z)p(z)dz \\&= \log \int_z q_{\phi}(z|x) \frac{p_{\theta}(x|z)p(z)}{q_{\phi}(z|x)} dz \\&= \log \mathbb{E}_{q_{\phi}(z|x)} \left[\frac{p_{\theta}(x|z)p(z)}{q_{\phi}(z|x)} \right] \\&\geq \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x|z)p(z)}{q_{\phi}(z|x)} \right] \\&= \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p(z)}{q_{\phi}(z|x)} \right] \\&= \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) || p(z))\end{aligned}$$

4th line: Jensen's inequality as applied to probability distributions. When f is concave, $f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$.

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$$\begin{array}{c} \vdots \\ \left. \begin{array}{c} \text{evidence} := \log p(x; \theta) \\ \text{ELBO} := \log E_{z \sim q} \left[\frac{p(x, z; \theta)}{q(z)} \right] \end{array} \right\} \text{KL}(q(z) \parallel p(z | x; \theta)) \\ \vdots \end{array}$$

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$$q_{\phi}(x, z)$$

$$\max_{\phi, \theta} \mathcal{L}_{\text{ELBO}} = \mathbb{E}_{p_D(x)} [\mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) || p(z))]$$

We can rewrite the objective as

$$\log \int_z p_{\theta}(x|z)p(z)dz = \log \int_z \frac{q_{\phi}(z|x)}{q_{\phi}(z|x)} p_{\theta}(x|z)p(z)dz$$

$$= \log \int_z q_{\phi}(z|x) \frac{p_{\theta}(x|z)p(z)}{q_{\phi}(z|x)} dz$$

$$= \log \mathbb{E}_{q_{\phi}(z|x)} \left[\frac{p_{\theta}(x|z)p(z)}{q_{\phi}(z|x)} \right]$$

$$= \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p(z)}{q_{\phi}(z|x)} \right]$$

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1. Expectation $\mathbb{E}_D[f(x)] = \frac{1}{N} \sum f(x_i) \quad \forall x_i \in D$

Using **enough samples** we can compute both of the expectations in the equation.

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2. Log of the decoder distribution

We assume a distribution over output like MVG or MVB to compute this analytically.

(Recall binary MNIST example)
(Remember log and exp cancels each other out)

$$\log p_\theta(x|z) = \mathcal{N}(x; \mu, \sigma^2 \mathbf{I} = 1)$$
$$\log p_\theta(x|z) \approx \text{MSE}(x, \mu)$$
$$\log p_\theta(x|z) \approx \text{BCE}(x, \text{out})$$

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3. KL divergence

We assume a Gaussian distribution over $p(z)$ & $q(z|x)$

Can again be computed analytically.

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KL divergence

We assume a Gaussian distribution over $p(z)$ [zero mean, std=1] & $q(z|x)$ [mean finite, variance finite]

Again our KL term can again be computed analytically.

$$q(z) = \mathcal{N}(z; \mu_1, \Sigma_1)$$

$$p(z) = \mathcal{N}(z; \mu_2, \Sigma_2)$$

$$\text{KL}(q(z) || p(z)) = \frac{1}{2} \left[\log \frac{|\Sigma_2|}{|\Sigma_1|} - d + \text{tr}\{\Sigma_2^{-1} \Sigma_1\} + (\mu_2 - \mu_1)^T \Sigma_2^{-1} (\mu_2 - \mu_1) \right]$$

Variational AutoEncoders (VAEs)

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KL divergence

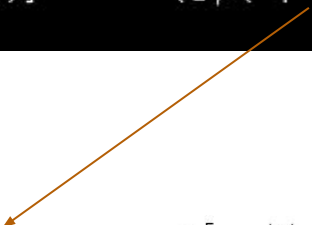
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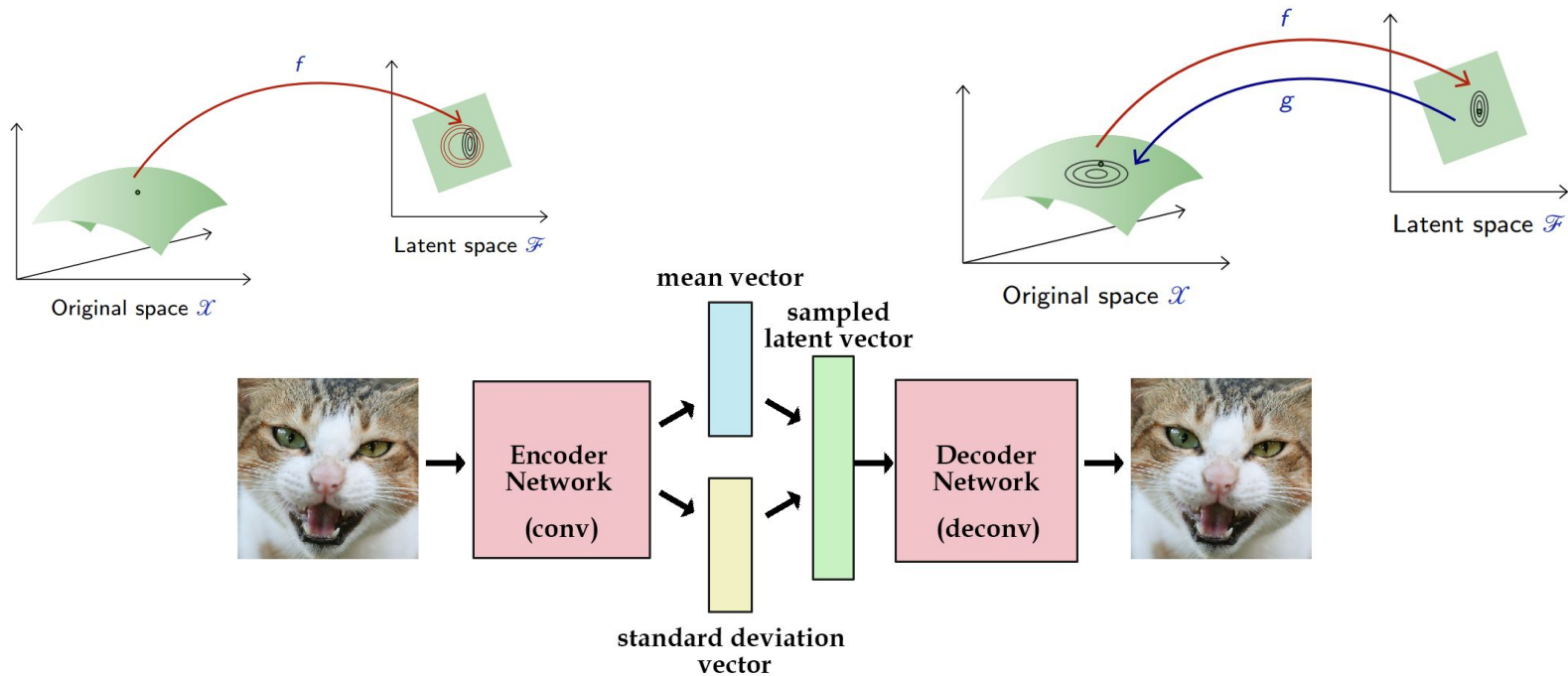
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$$\begin{aligned} \text{KL}(q_\phi(z|x) \parallel p(z)) &= \frac{1}{2} \left[\log\left(\frac{|I|}{|\Sigma|}\right) - d + \text{tr}\{I^{-1}\Sigma\} + (0 - \mu)^T I^{-1} (0 - \mu) \right] \\ &= \frac{1}{2} \left[-\log(|\Sigma|) - d + \text{tr}\{\Sigma\} + \mu^T \mu \right] \\ &= \frac{1}{2} \left[-\log\left(\prod_i \sigma_i^2\right) - d + \sum_i \sigma_i^2 + \sum_i \mu_i^2 \right] \\ &= \frac{1}{2} \left[-\sum_i \log(\sigma_i^2) - d + \sum_i \sigma_i^2 + \sum_i \mu_i^2 \right] \end{aligned}$$

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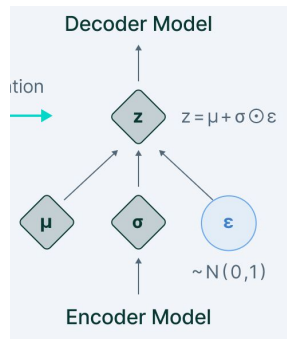
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```
def encode(self, x):
    output = self.encoder(x)
    mu, log_var = output[:, 0], output[:, 1]
    return mu, log_var
```

```
def decode(self, z):
    mu = self.decoder(z)
    return mu, mu.new_zeros(mu.size())
```

```
def sample_gaussian(param):
    mean, log_var = param
    std = log_var.mul(0.5).exp()
    return torch.randn(mean.size()) * std + mean
```



```
def log_p_gaussian(x, param):
    mean, log_var, x = param[0], param[1], x
    var = log_var.exp()
    return -0.5 * (((x - mean).pow(2) / var) + log_var +
    math.log(2 * math.pi)).sum(1)
```

```
def dkl_gaussians(param_a, param_b):
    mean_a, log_var_a = param_a[0], param_a[1]
    mean_b, log_var_b = param_b[0], param_b[1]
    var_a = log_var_a.exp()
    var_b = log_var_b.exp()
    return 0.5 * ( log_var_b - log_var_a - 1 + (mean_a -
    mean_b).pow(2) / var_b + var_a / var_b ).sum(1)
```

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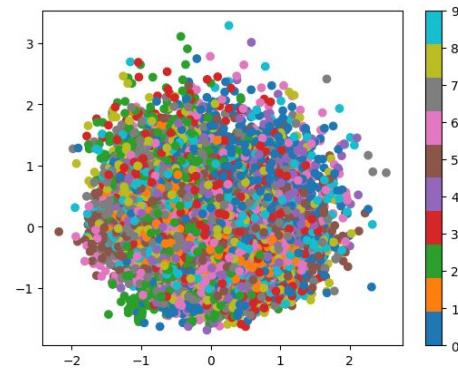
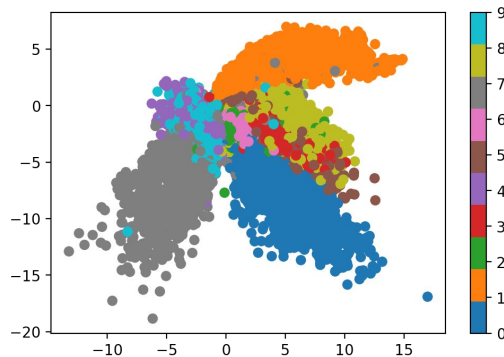
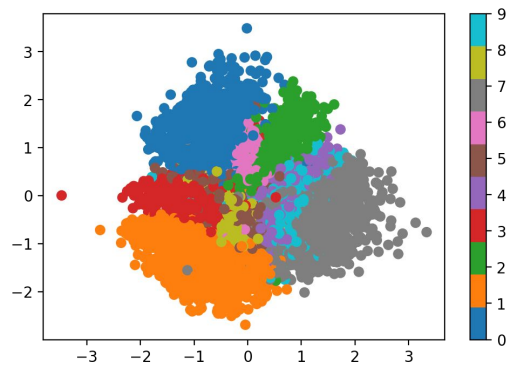
```
for x in train_input.split(args.batch_size):
    param_q_Z_given_x = model.encode(x)
    z = sample_gaussian(param_q_Z_given_x)
    param_p_X_given_z = model.decode(z)

    log_p_x_given_z = log_p_gaussian(x, param_p_X_given_z)
    dkl_q_Z_given_x_from_p_Z = dkl_gaussians(param_q_Z_given_x, param_p_Z)

    loss = -(log_p_x_given_z - dkl_q_Z_given_x_from_p_Z).mean()

    optimizer.zero_grad()
    loss.backward()
    optimizer.step()
```

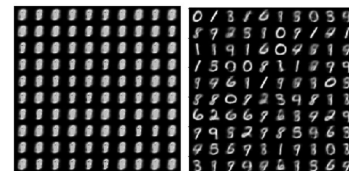

VAE and variants



Prior Variants: CVAE, GMM-VAE, S-VAE, VQ-VAE

Posterior variants: β VAE, InfoVAE, WAE

Note: auto-encoding can be done for sequential data also!



(a) 0 BPI

(b) 5 BPI



(c) 10 BPI

(d) 15 BPI